

Systems Under Pressure

Shared Group Investigation Notebook

DRIVING QUESTION

How do we know when a system is actually in danger?

GROUP DETAILS

GROUP NAME _____

MEMBERS _____

INVESTIGATION TOPIC _____

CATEGORY _____

ESSENTIAL QUESTION _____

HOW THIS NOTEBOOK WORKS

- One shared investigation journal per group.
- Individual handwritten extensions are separate and shown during vlogs.
- Three informal vlogs: sequences, exponentials, polynomials.
- No daily vlogs. No required 150-word reflections.

CORE ROUTINES

01 SYSTEM -> VARIABLES -> MODEL -> CRITIQUE

02 THINK FIRST -> SPARKY -> VERIFY -> DECIDE

03 LOCAL CHECK vs GLOBAL CHECK

04 WHAT THE MODEL REVEALS / HIDES

05 ONE REJECTED MODEL

THE WORK OF THIS NOTEBOOK

Graphs, equations, calculations, annotations, revision, evidence, and one rejected model.

Use this page for free exploration: what do we know, what surprises us, what makes this urgent, and what investigation question should we draft?

What We Know	What Surprises Us	Draft Question
List first observations before choosing a model.	Name what feels confusing, urgent, or worth investigating.	Write a focused group investigation question.

Use this page to identify who is affected, what they experience, what variables they would track, and where perspectives conflict.

SYSTEM	VARIABLES	MODEL	CRITIQUE
Stakeholders Who is affected, and how?	Variables What would each person measure or care about?	Conflict Where do perspectives disagree?	
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Variables + Hidden Systems

Use this page to map connected variables, possible hidden pressures, measurable evidence, and at least one confounding variable.

SYSTEM	VARIABLES	MODEL	CRITIQUE
System Map What connected pressures are visible?	Measurable What can be measured with real data?	Hidden Pressure What might be missing or confounding?	
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Graph Observations

Use this page to annotate trends, thresholds, anomalies, and intervals before choosing equations.

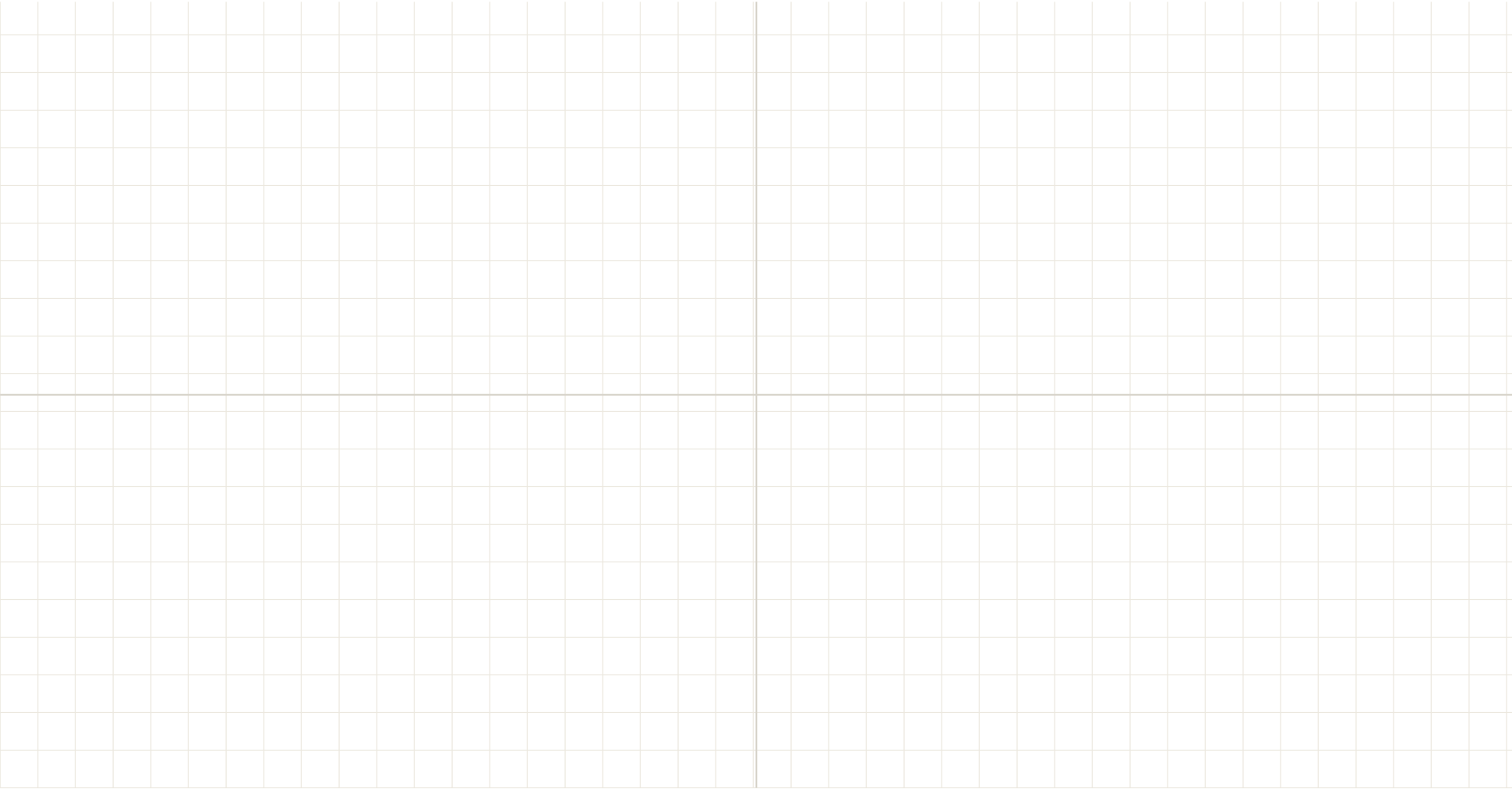
SYSTEM	VARIABLES	MODEL	CRITIQUE
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Graph First
What feature matters most?

Interval
Where does the pattern change?

Evidence
Which part supports your claim?

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Local vs Global Behavior

Use this page to compare a short interval with a longer interval and notice model failure.

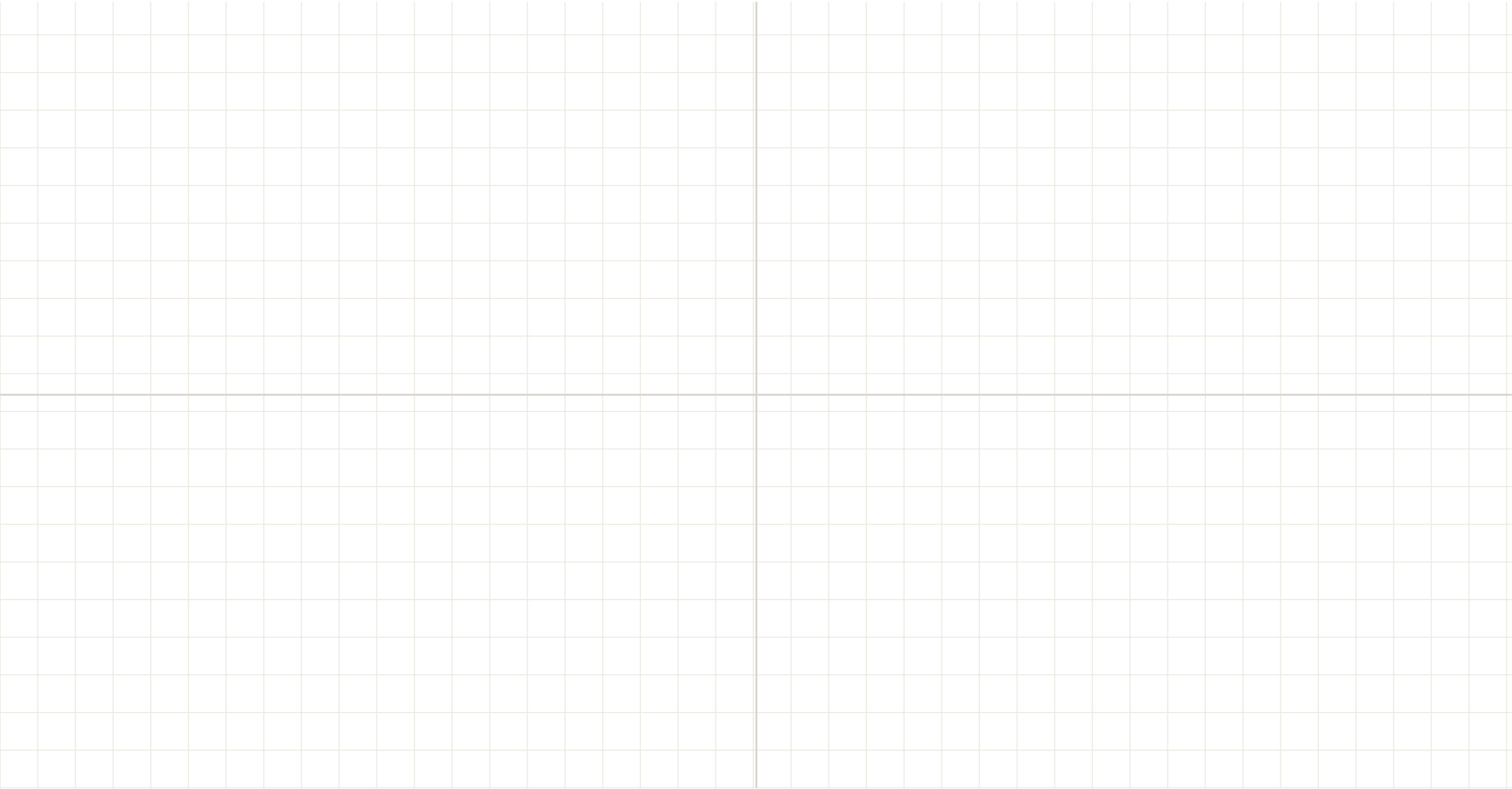
SYSTEM	VARIABLES	MODEL	CRITIQUE
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Zoom In
What seems true here?

Zoom Out
Does it still make sense globally?

Bad Extrapolation
Where did the model stop being meaningful?

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Sequences + Series Modeling

Use this page to model repeated change, accumulation, prediction, and where the pattern breaks.

SYSTEM	VARIABLES	MODEL	CRITIQUE
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Recursive
How does each term depend on the one before it?

recursive $a_n = r \cdot a_{n-1} + d$

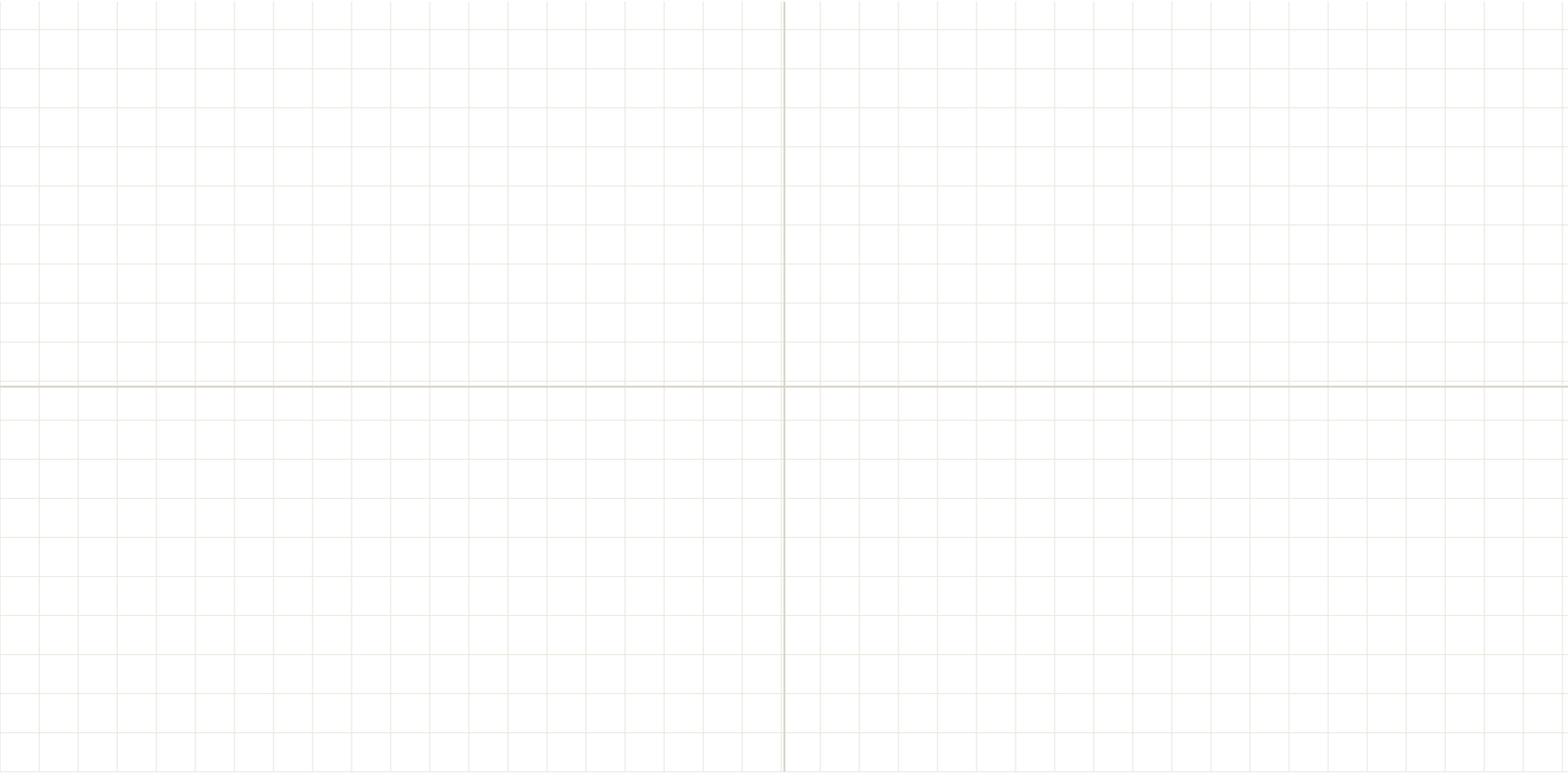
Explicit
Can you predict term n directly?

explicit $a_n = a_1 r^{n-1}$

Accumulation
What does the partial sum represent?

series $S_n = a_1 + a_2 + \dots + a_n$

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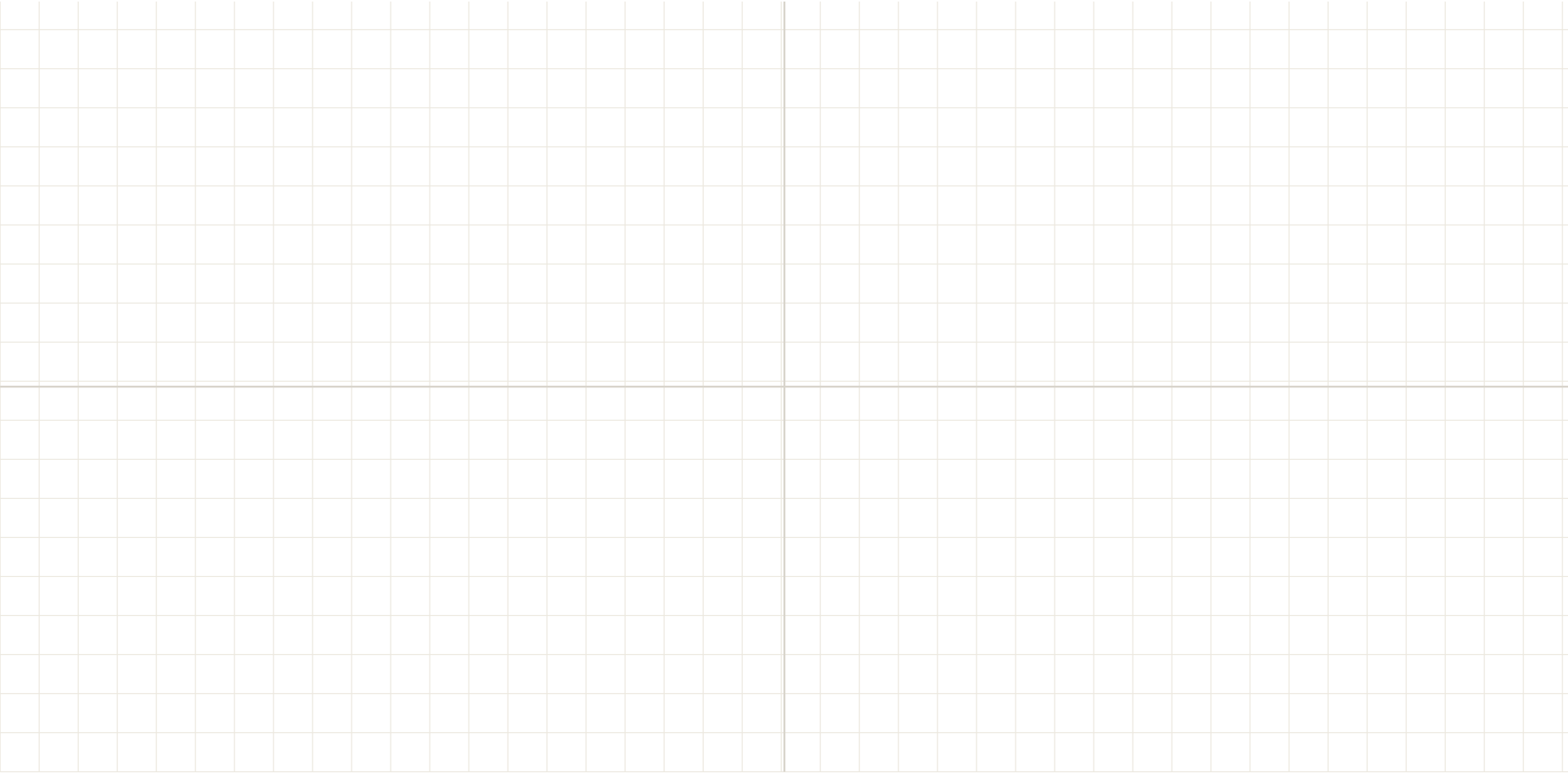
Exponentials + Thresholds

Use this page to test ratios, solve thresholds, interpret parameters, and name where exponential reasoning fails.

SYSTEM	VARIABLES	MODEL	CRITIQUE
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Ratio Test Are the ratios stable enough? model $f(x) = a \cdot b^x$	Threshold What value matters in the system? solve $\log_b(\text{threshold}/a) = x$	Failure Where does growth/decay become unrealistic? meaning a = initial value, b = growth/decay factor
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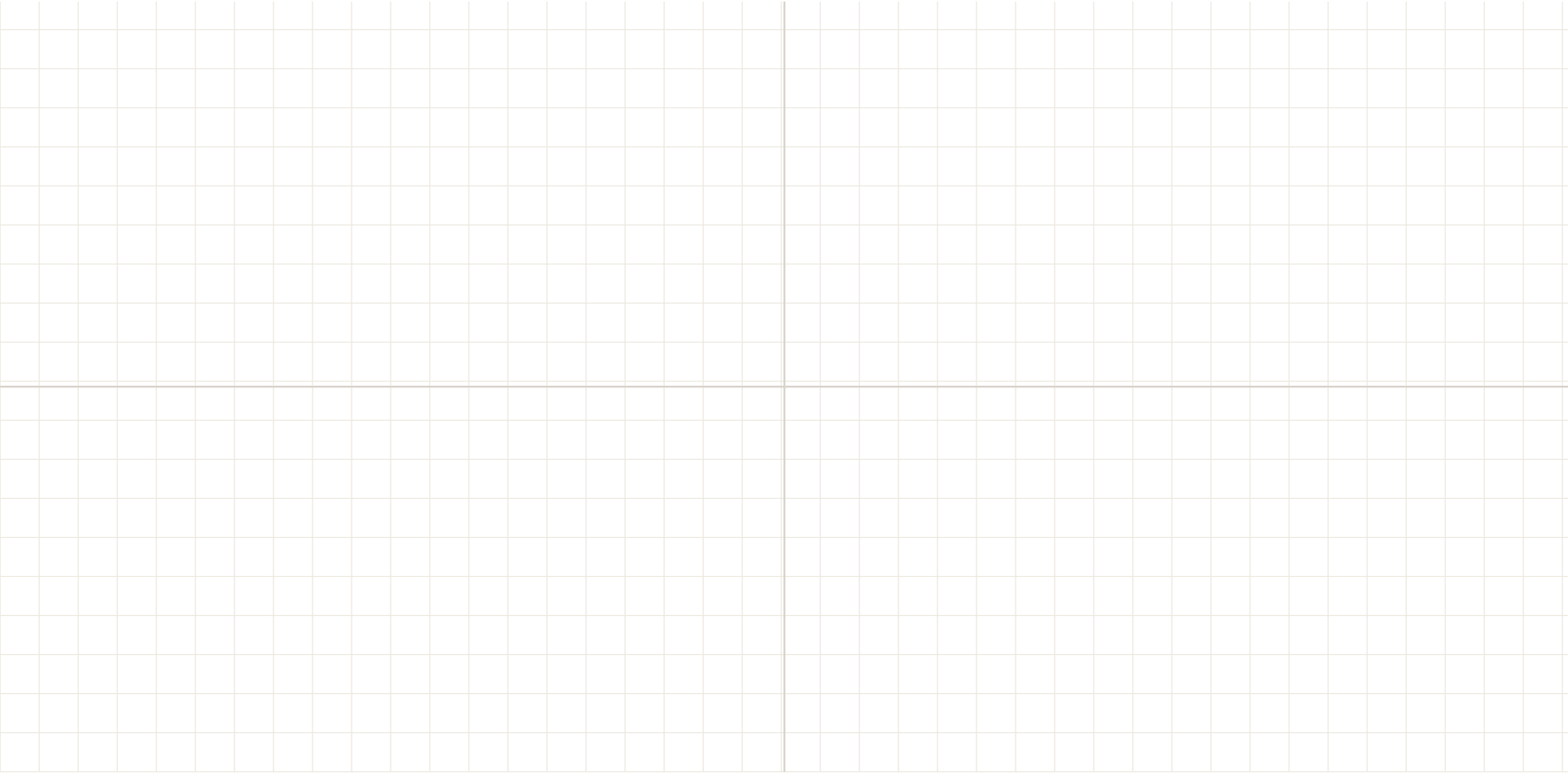
Polynomial Behavior

Use this page to interpret local shape, turning points, zeros, thresholds, multiplicity, and end behavior realism.

SYSTEM	VARIABLES	MODEL	CRITIQUE
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Local Model What interval justifies this shape? factored $P(t) = a(t-r_1)(t-r_2)^2...$	Turning Point What does the reversal mean physically? zero $P(t)=0$	End Behavior Where does the polynomial stop making sense? feature multiplicity: cross or bounce?
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Use this page to compare models by what they predict, reveal, hide, assume, and fail to explain.

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Confounding Variables

Use this page to separate visible correlation from possible causes and alternative explanations.

SYSTEM	VARIABLES	MODEL	CRITIQUE
Related What evidence suggests connection?	Not Proof What else could explain it?	Sentence Frame These variables appear related because ____, but this does not prove causation because ____.	
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Use this page only after your group thinks first: record what Sparky suggested, what you verified, and what you rejected.

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Use this page to show how evidence, critique, or model failure changed your group's interpretation.

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Use this page to make a defensible claim with graph evidence, calculation evidence, rejected-model evidence, and uncertainty.

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Use this page to prepare a mathematical argument: strongest evidence, biggest limitation, rejected model, and final defense.

Explain Every graph and equation verbally.	Defend Mathematical, realism, assumption, interval, counterargument tests.	Close Not a summary. A mathematical argument.
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Use this page for last annotations, slide notes, and questions your group must be ready to answer.

Ready All members can explain the math.	Artifacts Notebook, vlogs, extensions, and sources are ready.	Attend Presentation day is during class on Tuesday, June 9.
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